



The cutoff frequencies ω_1 and ω_2 (in rad/sec) are

14. The voltage across the inductor (V_L) at resonance (in V) is

Prob. 4: Consider the circuit shown in Figure 3, and answer questions 15-17.

15. The current supplied by the source (I_1)
16. The voltage (V_0)
17. The complex power supplied by the source is

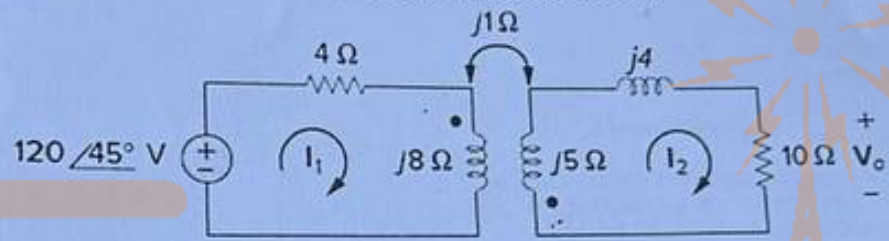


Figure 3

Prob. 5: Consider a balanced three phase Y-Y system has $V_{an} = 440/0^\circ$ V_{rms} in positive phase sequence and a load of $Z_y = 6-j8 \Omega$. If the line impedance is zero as shown in figure 4 and answer questions 18-22.

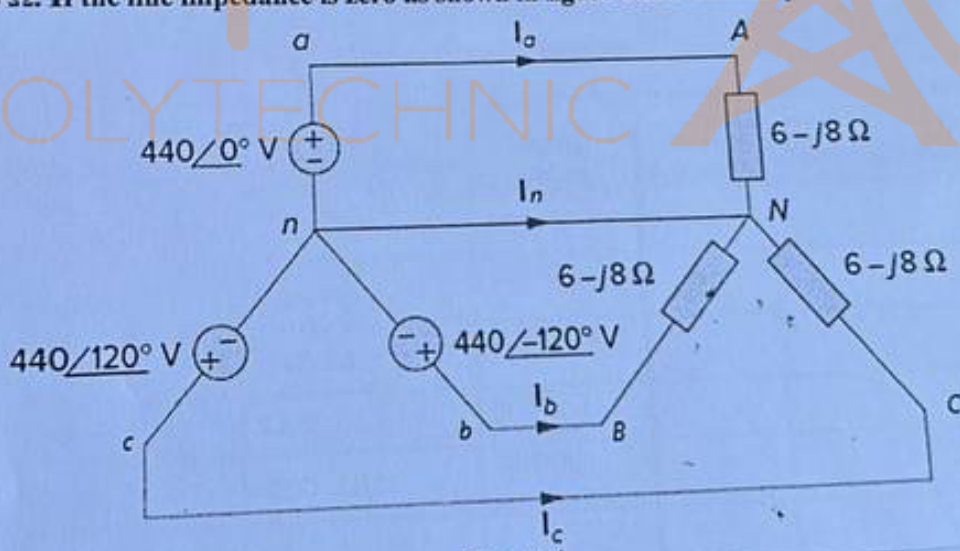


Figure 4

18. The voltage V_{AB} is:
19. The line current I_a is:
20. The phase current I_b is:
21. The total average power supplied by the source is:
22. The total complex power consumed by the load is:
23. The neutral line current is:

Good Luck



Write all the required equations (without simplification) to solve the following problems, then find the final answer.

Prob. 1: Consider the circuit shown in Figure 1 and answer questions 1-4.

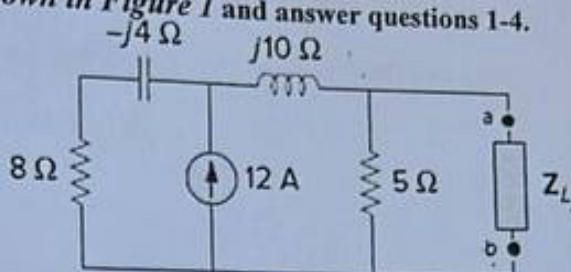


Figure 1

1. The Thevenin resistance with respect to terminals a-b is:
.....
2. The Thevenin voltage with respect to terminals a-b is:
.....
3. The load impedance Z_L that absorbs the maximum average power is.
.....
4. The maximum average power absorbed by Z_L is.
.....

Prob. 2: Two Loads are connected in parallel across a voltage source of $440\angle 0^\circ V_{rms}$ as shown in figure 2. Answer questions 5-10.

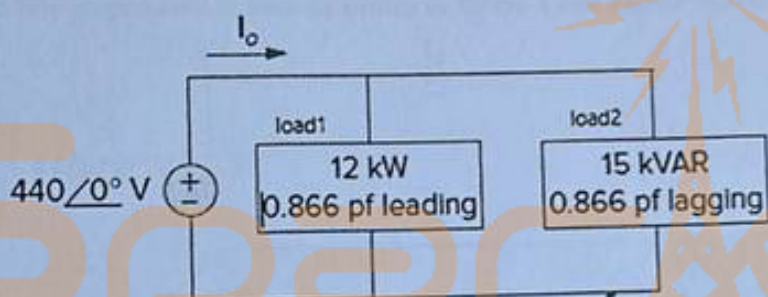


Figure 2

5. The current of load 1 is:
.....
6. The current of load 2 is:
.....
7. The complex power supplied by the source is:
.....
8. The power factor of the equivalent load is:
.....
9. The current supplied by the voltage source in rms is:
.....
10. The impedance of the equivalent load is:
.....

Prob. 3: Consider a series RLC circuit has $R = 5 \Omega$, $L = 4mH$, $C = 0.1\mu F$, and $V_s = 10\angle 0^\circ V_{rms}$ and answer questions 11-14.

11. The resonant frequency (ω_o) in (rad/sec) is:
.....
12. The quality factor (Q) is:
.....